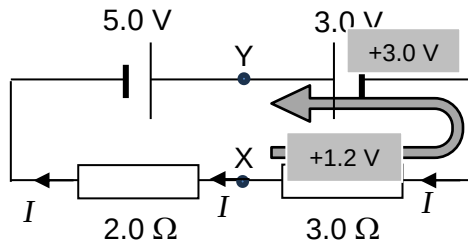


22 D Effective e.m.f. = 2.0 V

$$\text{p.d. across the } 3.0 \, \Omega \text{ resistor} = \frac{3.0}{2.0 + 3.0} (2.0) = 1.2 \, \text{V}$$

p.d. between X and Y = 1.2 + 3.0 = 4.2 V
(going from negative terminal to positive terminal of 3.0 V battery, potential increases by 3.0 V)



22 **A** Since Y and Z carry the same current of 3.0 A and are the same distance from X , the